

Automotive Power Train

□ To explain the principle of engine control system and transmission control systems

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5053A is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5053A.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Automobile Engineering
Course code	5053A
Course title	Automotive Power Train
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L:4, T:0, P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

13 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To explain the principle of engine control system and transmission control systems

Course outcomes

CO	Official outcome	Study level
CO1	14 Understanding Systems Identify various sensors and actuators used in	See official syllabus
CO2	15 Applying electronic engine control system.	See official syllabus
CO3	Outline the features of digital engine control system 15 Understanding Outline the integrated engine control system and	See official syllabus
CO4	14 Understanding automatic transmission control	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 2
CO2	CO2 3 2
CO3	CO3 3 2
CO4	CO4 2 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Summarize the concept of electronic engine control Systems	3	As specified
Module 2	system.	4	As specified
Module 3	Outline the features of digital engine control system	3	As specified
Module 4	control	3	As specified

MODULE 1

Summarize the concept of electronic engine control Systems

This module is presented from the official syllabus scope for Automotive Power Train. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize the concept of electronic engine control Systems
- Official duration: As specified
- Official topic entries captured: 3

- The full source excerpt is preserved below for coverage checking.

Outline the functions of engine and controls

Outline the functions of engine and controls is an official topic in Module 1 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the functions of engine and controls using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the functions of engine and controls should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the functions of engine and controls means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Outline the functions of engine and controls
 ഏകദേശം definition/meaning ഉൾപ്പെടെ. ഉൾപ്പെടെ ഉൾപ്പെടെ,
 steps, application ഉൾപ്പെടെ ഉൾപ്പെടെ. Technical terms English- ഉൾപ്പെടെ
 ഉൾപ്പെടെ.

Outline the terms related to engine performance

Outline the terms related to engine performance is an official topic in Module 1 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the terms related to engine performance using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the terms related to engine performance should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the terms related to engine performance means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Outline the terms related to engine performance
 definition/meaning .
 , steps, application . Technical terms
 English- .

Explain the mapping of engine. 3 Understanding M1. 04 Outline electronic fuel control system 4 Understanding The concept of electronic Engine control system - Engine functions and control - Parameters and variables - Input to controller - output from controller - Definition of engine performance terms - Power - torque - fuel consumption - Volumetric efficiency - Thermal efficiency - calibration, Engine mapping-Effect of air fuel ratio, spark timing and exhaust gas recirculation on performance - Electronic fuel control system - Engine control sequence - open loop control - closed loop control - closed loop operation - Idle speed control. Identify various sensors and actuators used in electronic engine control

Explain the mapping of engine. 3 Understanding M1. 04 Outline electronic fuel control system 4 Understanding The concept of electronic Engine control system - Engine functions and control - Parameters and variables - Input to controller - output from controller - Definition of engine performance terms - Power - torque - fuel consumption - Volumetric efficiency - Thermal efficiency - calibration, Engine mapping-Effect of air fuel ratio, spark timing and exhaust gas recirculation on performance - Electronic fuel control system - Engine control sequence - open loop control - closed loop control - closed loop operation - Idle speed control. Identify various sensors and actuators used in electronic engine control is an official topic in Module 1 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the mapping of engine. 3 Understanding M1. 04 Outline electronic fuel control system 4 Understanding The concept of electronic Engine control system - Engine functions and control - Parameters and variables - Input to controller - output from controller - Definition of engine performance terms - Power - torque - fuel consumption - Volumetric efficiency - Thermal efficiency - calibration, Engine mapping-Effect of air fuel ratio, spark timing and exhaust gas recirculation on performance - Electronic fuel control system - Engine control sequence - open loop control - closed loop control - closed loop operation - Idle speed control. Identify various sensors and actuators used in electronic engine control using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the mapping of engine. 3 understanding m1. 04 outline electronic fuel control system 4 understanding the concept of electronic engine control system - engine functions and control - parameters and variables - input to controller - output from controller - definition of engine performance terms - power - torque - fuel consumption - volumetric efficiency - thermal efficiency - calibration, engine mapping- effect of air fuel ratio, spark timing and exhaust gas recirculation on performance - electronic fuel control system - engine control sequence - open loop control - closed loop control - closed loop operation - idle speed control. identify various sensors and actuators used in electronic engine control should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the mapping of engine. 3 Understanding M1. 04 Outline electronic fuel control system 4 Understanding The concept of electronic Engine control system - Engine functions and control - Parameters and variables - Input to controller - output from controller - Definition of engine performance terms - Power - torque - fuel consumption - Volumetric efficiency - Thermal efficiency - calibration, Engine mapping-Effect of air fuel ratio, spark timing and exhaust gas recirculation on performance - Electronic fuel control system - Engine control sequence - open loop control - closed loop control - closed loop operation - Idle speed control. Identify various sensors and actuators used in electronic engine control means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain the mapping of engine. 3 Understanding M1. 04 Outline electronic fuel control system 4 Understanding The concept of electronic Engine control system - Engine functions and control - Parameters and variables - Input to controller - output from controller - Definition of

engine performance terms - Power - torque - fuel consumption - Volumetric efficiency - Thermal efficiency - calibration, Engine mapping-Effect of air fuel ratio, spark timing and exhaust gas recirculation on performance - Electronic fuel control system - Engine control sequence - open loop control - closed loop control - closed loop operation - Idle speed control. Identify various sensors and actuators used in electronic engine control **മലയാളം** definition/meaning **മലയാളം**. **മലയാളം** **മലയാളം** **മലയാളം**, steps, application **മലയാളം** **മലയാളം** **മലയാളം**. Technical terms English-**മലയാളം** **മലയാളം** **മലയാളം**.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Summarize the concept of electronic engine control Systems

M1.01	Outline the functions of engine and controls	3
Understanding		
M1.02	Outline the terms related to engine performance	4
Understanding		
M1.03	Explain the mapping of engine.	3
Understanding		
M1. 04	Outline electronic fuel control system	4
Understanding		

Contents:

The concept of electronic Engine control system - Engine functions and control - Parameters and variables - Input to controller - output from controller - Definition of engine performance terms - Power - torque - fuel consumption - Volumetric efficiency - Thermal efficiency - calibration, Engine mapping-Effect of air fuel ratio, spark timing and exhaust gas recirculation on performance - Electronic fuel control system - Engine control sequence - open loop control - closed loop control - closed loop operation - Idle speed control.

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam

MODULE 2

system.

This module is presented from the official syllabus scope for Automotive Power Train. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: system.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Outline the engine Variables

Outline the engine Variables is an official topic in Module 2 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the engine Variables using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the engine variables should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the engine Variables means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഓൗട്ലൈൻ എൻജിൻ വേരിയബിൾസ്
ഓൗട്ട്ലൈൻ definition/meaning ഓൗട്ട്ലൈൻ. ഓൗട്ട്ലൈൻ ഓൗട്ട്ലൈൻ steps,
application ഓൗട്ട്ലൈൻ ഓൗട്ട്ലൈൻ. Technical terms English-ഓൗട്ട്ലൈൻ
ഓൗട്ട്ലൈൻ.

Illustrate the working of various sensors

Illustrate the working of various sensors is an official topic in Module 2 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the working of various sensors using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the working of various sensors should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the working of various sensors means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഈ Module-ൽ വിവിധ സെൻസറുകളുടെ പ്രവർത്തനം വിശദീകരിക്കുക. definition/meaning, steps, application എന്നിവ വിശദീകരിക്കുക. Technical terms English-ൽ വിശദീകരിക്കുക.

Illustrate the working of various actuators

Illustrate the working of various actuators is an official topic in Module 2 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the working of various actuators using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the working of various actuators should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the working of various actuators means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഈ Module-ൽ Illustrate the working of various actuators എന്നാണ് ഉദ്ദേശ്യം. ഈ Module-ൽ definition/meaning, steps, application എന്നിവയെക്കുറിച്ച് വിവരിക്കുക. Technical terms English-ൽ എഴുതുക.

Identify various actuators and actuators 1 Applying Typical electronic Engine Control System. Variables to be measured, Sensors and actuators- Temperature sensor- Pressure sensor-Mass air flow sensor-Oxygen sensor- Position sensor-Position/speed sensor-Knock sensor- Automotive Engine Control Actuators-Solenoid-switches-relays-motors-Fuel injection- Exhaust gas recirculation actuators-Variable valve timing.

Identify various actuators and actuators 1 Applying Typical electronic Engine Control System. Variables to be measured, Sensors and actuators- Temperature sensor- Pressure sensor-Mass air flow sensor-Oxygen sensor- Position sensor- Position/speed sensor-Knock sensor- Automotive Engine Control Actuators- Solenoid-switches-relays-motors-Fuel injection- Exhaust gas recirculation actuators-Variable valve timing. is an official topic in Module 2 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify various actuators and actuators 1 Applying Typical electronic Engine Control System. Variables to be measured, Sensors and actuators- Temperature sensor- Pressure sensor-Mass air flow sensor-Oxygen sensor- Position sensor-Position/speed sensor-Knock sensor- Automotive Engine Control Actuators-Solenoid-switches-relays-motors-Fuel injection- Exhaust gas recirculation actuators-Variable valve timing. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify various actuators and actuators 1 applying typical electronic engine control system. variables to be measured, sensors and actuators- temperature sensor- pressure sensor-mass air flow sensor-oxygen sensor- position sensor-position/speed sensor-knock sensor- automotive engine control actuators- solenoid-switches-relays-motors-fuel injection- exhaust gas recirculation actuators-

variable valve timing. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify various actuators and actuators 1 Applying Typical electronic Engine Control System. Variables to be measured, Sensors and actuators- Temperature sensor- Pressure sensor- Mass air flow sensor-Oxygen sensor- Position sensor-Position/speed sensor-Knock sensor- Automotive Engine Control Actuators-Solenoid-switches-relays-motors-Fuel injection- Exhaust gas recirculation actuators-Variable valve timing. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-Identify various actuators and actuators 1 Applying Typical electronic Engine Control System. Variables to be measured, Sensors and actuators- Temperature sensor- Pressure sensor-Mass air flow sensor- Oxygen sensor- Position sensor-Position/speed sensor-Knock sensor- Automotive Engine Control Actuators-Solenoid-switches-relays-motors-Fuel injection- Exhaust gas recirculation actuators-Variable valve timing. definition/meaning steps, application Technical terms English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

system.

M2.01	Outline the engine Variables Understanding	4
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M2.02	Illustrate the working of various sensors Understanding	7
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M2.03	Illustrate the working of various actuators Understanding	3
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M2.04	Identify various actuators and actuators Applying	1
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	Series Test – I	1
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Contents:

Typical electronic Engine Control System. Variables to be measured, Sensors and

actuators- Temperature sensor- Pressure sensor-Mass air flow sensor-Oxygen sensor-

Position sensor-Position/speed sensor-Knock sensor-

Automotive Engine Control Actuators-Solenoid-switches-relays-motors-Fuel injection-

Exhaust gas recirculation actuators-Variable valve timing.

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Outline the features of digital engine control system

This module is presented from the official syllabus scope for Automotive Power Train. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Outline the features of digital engine control system
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

5 Understanding control systems.

5 Understanding control systems. is an official topic in Module 3 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding control systems. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding control systems. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding control systems. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-5 Understanding control systems.

നിയന്ത്രണ സംവിധാനം നിയന്ത്രണ definition/meaning നിയന്ത്രണ സംവിധാനം. നിയന്ത്രണ സംവിധാനം നിയന്ത്രണ, steps, application നിയന്ത്രണ സംവിധാനം നിയന്ത്രണ. Technical terms English-നിയന്ത്രണ സംവിധാനം നിയന്ത്രണ.

Explain various modes for fuel control

Explain various modes for fuel control is an official topic in Module 3 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain various modes for fuel control using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain various modes for fuel control should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain various modes for fuel control means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain various modes for fuel control

Explain various modes for fuel control. definition/meaning, steps, application, Technical terms English-Malayalam.

Explain Electronic ignition control 5 Understanding Digital power train control systems - features - Components - Digital engine control system diagram - Modes for fuel control - Control during Engine crank - Engine Warm up - After the engine warm up - open loop - closed loop - Acceleration enrichment - Deceleration leaning - Idle speed control - EGR Control - Variable valve timing control - Electronic ignition control-Distributor less Ignition control. Outline the integrated engine control system and automatic transmission

Explain Electronic ignition control 5 Understanding Digital power train control systems - features - Components - Digital engine control system diagram - Modes for fuel control - Control during Engine crank - Engine Warm up - After the engine warm up - open loop - closed loop - Acceleration enrichment - Deceleration leaning - Idle speed control - EGR Control - Variable valve timing control - Electronic ignition control-Distributor less Ignition control. Outline the integrated engine control system and automatic transmission is an official topic in Module 3 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Electronic ignition control 5 Understanding Digital power train control systems - features - Components - Digital engine control system diagram - Modes for fuel control - Control during Engine crank - Engine Warm up - After the engine warm up - open loop - closed loop - Acceleration enrichment - Deceleration leaning - Idle speed control - EGR Control - Variable valve timing control - Electronic ignition control-Distributor less Ignition control. Outline the integrated engine control system and automatic transmission using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain electronic ignition control 5 understanding digital power train control systems - features - components - digital engine control system diagram - modes for fuel control - control during engine crank - engine warm up - after the engine warm up - open loop - closed loop - acceleration enrichment - deceleration leaning - idle speed control - egr control - variable valve timing control - electronic ignition control- distributor less ignition control. outline the integrated engine control system and automatic transmission should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Electronic ignition control 5 Understanding Digital power train control systems - features - Components - Digital engine control system diagram - Modes for fuel control - Control during Engine crank - Engine Warm up - After the engine warm up - open loop - closed loop - Acceleration enrichment - Deceleration leaning - Idle speed control - EGR Control - Variable valve timing control - Electronic ignition control-Distributor less Ignition control. Outline the integrated engine control system and automatic transmission means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain Electronic ignition control 5 Understanding Digital power train control systems - features - Components - Digital engine control system diagram - Modes for fuel control - Control during Engine crank - Engine Warm up - After the engine warm up - open loop - closed loop - Acceleration enrichment - Deceleration leaning - Idle speed control - EGR Control - Variable valve timing control - Electronic ignition control-Distributor less Ignition control. Outline the integrated engine control system and automatic transmission.
 പരീക്ഷയിൽ എഴുതേണ്ടത് definition/meaning ഉൾപ്പെടെയുള്ളവയാണ്. പരീക്ഷയിൽ എഴുതേണ്ടത് steps, application ഉൾപ്പെടെയുള്ളവയാണ്. Technical terms English-ൽ എഴുതേണ്ടത് ഉൾപ്പെടെയുള്ളവ.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Outline the features of digital engine control system
Illustrate the features of digital power train

M3.01		5
Understanding	control systems.	

M3.02	Explain various modes for fuel control	5
Understanding		

M3.03	Explain Electronic ignition control	5
Understanding		

Contents:

Digital power train control systems - features - Components - Digital engine control system diagram - Modes for fuel control - Control during Engine crank - Engine Warm up - After the engine warm up - open loop - closed loop - Acceleration enrichment - Deceleration leaning - Idle speed control - EGR Control - Variable valve timing control - Electronic ignition control-Distributor less Ignition control.

Outline the integrated engine control system and automatic transmission

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

control

This module is presented from the official syllabus scope for Automotive Power Train.
Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: control
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Explain integrated engine control systems 4 Understanding Explain improvements in Electronic engine

Explain integrated engine control systems 4 Understanding Explain improvements in Electronic engine is an official topic in Module 4 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain integrated engine control systems 4 Understanding Explain improvements in Electronic engine using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain integrated engine control systems 4 understanding explain improvements in electronic engine should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain integrated engine control systems 4 Understanding Explain improvements in Electronic engine means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്നിട്ടുള്ളതുകൊണ്ട് Explain integrated engine control systems 4 Understanding Explain improvements in Electronic engine എന്തുകൊണ്ട് എന്തുകൊണ്ട് definition/meaning എന്തുകൊണ്ട് എന്തുകൊണ്ട്. എന്തുകൊണ്ട് എന്തുകൊണ്ട് steps, application എന്തുകൊണ്ട് എന്തുകൊണ്ട് എന്തുകൊണ്ട്. Technical terms English-എന്നിട്ടുള്ളതുകൊണ്ട് എന്തുകൊണ്ട് എന്തുകൊണ്ട്.

5 Understanding control system Explain automatic transmission and traction

5 Understanding control system Explain automatic transmission and traction is an official topic in Module 4 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding control system Explain automatic transmission and traction using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding control system explain automatic transmission and traction should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding control system Explain automatic transmission and traction means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-5 Understanding control system Explain automatic transmission and traction ഓട്ടോമേറ്റഡ് ട്രാക്ഷൻ definition/meaning ഓട്ടോമേറ്റഡ് ട്രാക്ഷൻ. ഓട്ടോമേറ്റഡ് ട്രാക്ഷൻ, steps, application ഓട്ടോമേറ്റഡ് ട്രാക്ഷൻ. Technical terms English-ഓട്ടോമേറ്റഡ് ട്രാക്ഷൻ.

**5 Understanding control Integrated engine control system - Secondary air Management - Evaporative emission canister purge - Automatic system adjustment - System diagnosis - Improvements in Electronic Engine Control system - Oxygen sensor improvements - Heated exhaust gas oxygen sensor - Fuel injection Timing - Transmission control - Automatic transmission control - Torque convertor lock up control - Differential and traction control Text / Reference: T/R Book Title/Author T1 U.Kienck, and L. Nielsen, Automotive Control Systems, SAE and Springer-Verlag 2000 LjuboVlacic, Michel Parent, Fumio Harashima, "Intelligent Vehicle Technologies", R1 Butterworth-Heinemann publications, Oxford, 2001 Crouse, W.H & Anglin, D.L,"Automotive mechanics", Intl.Student edition, R2 9thedition, TMH, New Delhi, 2002. William B. Ribbens-Understanding Automotive Electronics, 5th edition, Butter worth R3 Heinemann Woburn 1998. R4 Bosh, "Automotive Handbook", 6th edition, SAE, 2004 Online Resources: Sl.No Website Link
<https://www.electronicsforu.com/technology-trends/tech-focus/automobile-1> industry-sensor 2 <https://www.youtube.com/watch?v=jyQuRgQHGCK> 3 <https://www.youtube.com/watch?v=8e4QB1DYYJU> 4 <https://www.youtube.com/watch?v=FpGGpgSEU94> 5 <https://www.youtube.com/watch?v=PFNe7NyJeb0>**

5 Understanding control Integrated engine control system - Secondary air Management - Evaporative emission canister purge - Automatic system adjustment - System diagnosis - Improvements in Electronic Engine Control system - Oxygen sensor improvements - Heated exhaust gas oxygen sensor - Fuel injection Timing - Transmission control - Automatic transmission control - Torque convertor lock up control - Differential and traction control Text / Reference: T/R Book Title/Author T1 U.Kienck, and L. Nielsen, Automotive Control Systems, SAE and Springer-Verlag 2000 LjuboVlacic, Michel Parent, Fumio Harashima, "Intelligent Vehicle Technologies", R1 Butterworth-Heinemann publications, Oxford, 2001 Crouse, W.H & Anglin, D.L,"Automotive mechanics", Intl.Student edition, R2 9thedition, TMH, New Delhi, 2002. William B. Ribbens-Understanding Automotive Electronics, 5th edition, Butter worth R3 Heinemann Woburn 1998. R4 Bosh, "Automotive Handbook", 6th edition, SAE, 2004 Online Resources: Sl.No Website Link <https://www.electronicsforu.com/technology-trends/tech-focus/automobile-1> industry-sensor 2 <https://www.youtube.com/watch?v=jyQuRgQHGCK> 3 <https://www.youtube.com/watch?v=8e4QB1DYYJU> 4 <https://www.youtube.com/watch?v=FpGGpgSEU94> 5 <https://www.youtube.com/watch?v=PFNe7NyJeb0> is an official topic in Module 4 of Automotive Power Train. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding control Integrated engine control system - Secondary air Management - Evaporative emission canister purge - Automatic system adjustment - System diagnosis - Improvements in Electronic Engine Control system - Oxygen sensor improvements - Heated exhaust gas oxygen sensor - Fuel injection Timing - Transmission control - Automatic transmission control - Torque convertor lock up control - Differential and traction control Text / Reference: T/R Book Title/Author T1 U.Kienck, and L. Nielsen, Automotive Control Systems, SAE and Springer-Verlag 2000 LjuboVlacic, Michel Parent, Fumio Harashima, "Intelligent Vehicle Technologies", R1 Butterworth-Heinemann publications, Oxford, 2001 Crouse, W.H & Anglin, D.L,"Automotive mechanics", Intl.Student edition, R2 9thedition,TMH,NewDelhi,2002. William B. Ribbens-Understanding Automotive Electronics, 5th edition, Butter worth R3 Heinemann Woburn 1998. R4 Bosh, "Automotive Handbook", 6th edition, SAE,2004 Online Resources: SI.No Website Link <https://www.electronicsforu.com/technology-trends/tech-focus/automobile-1> industry-sensor 2 <https://www.youtube.com/watch?v=jyQuRgQHGCK> 3 <https://www.youtube.com/watch?v=8e4QB1DYYJU> 4 <https://www.youtube.com/watch?v=FpGGpgSEU94> 5 <https://www.youtube.com/watch?v=PFNe7NyJeb0> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding control integrated engine control system - secondary air management - evaporative emission canister purge - automatic system adjustment - system diagnosis - improvements in electronic engine control system - oxygen sensor improvements - heated exhaust gas oxygen sensor - fuel injection timing - transmission control - automatic transmission control - torque convertor lock up control - differential and traction control text / reference: t/r book title/author t1 u.kienck, and l. nielsen, automotive control systems, sae and springer-verlag 2000 ljubovlacic, michel parent, fumio harashima, "intelligent vehicle technologies", r1 butterworth-heinemann publications, oxford, 2001 crouse, w.h & anglin, d.l,"automotive mechanics", intl.student edition, r2 9thedition,tmh,newdelhi,2002. william b. ribbens-

understanding automotive electronics, 5th edition, butter worth r3 heinemann woburn 1998. r4 bosh, "automotive handbook", 6th edition, sae,2004 online resources: sl.no website link <https://www.electronicsforu.com/technology-trends/tech-focus/automobile-1> industry-sensor 2 <https://www.youtube.com/watch?v=jyqurgqhgck> 3 <https://www.youtube.com/watch?v=8e4qb1dyyju> 4 <https://www.youtube.com/watch?v=fpggpgseu94> 5 <https://www.youtube.com/watch?v=pfne7nyjeb0> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding control Integrated engine control system - Secondary air Management - Evaporative emission canister purge - Automatic system adjustment - System diagnosis - Improvements in Electronic Engine Control system - Oxygen sensor improvements - Heated exhaust gas oxygen sensor - Fuel injection Timing - Transmission control - Automatic transmission control - Torque convertor lock up control - Differential and traction control Text / Reference: T/R Book Title/Author T1 U.Kienck, and L. Nielsen, Automotive Control Systems, SAE and Springer-Verlag 2000 LjuboVlacic, Michel Parent, Fumio Harashima, "Intelligent Vehicle Technologies", R1 Butterworth-Heinemann publications, Oxford, 2001 Crouse, W.H & Anglin, D.L,"Automotive mechanics", Intl.Student edition, R2 9thedition,TMH,NewDelhi,2002. William B. Ribbens- Understanding Automotive Electronics, 5th edition, Butter worth R3 Heinemann Woburn 1998. R4 Bosh, "Automotive Handbook", 6th edition, SAE,2004 Online Resources: SI.No Website Link https://www.electronicsforu.com/technology-trends/tech-focus/automobile-1 industry-sensor 2 https://www.youtube.com/watch?v=jyQuRgQHGCk 3 https://www.youtube.com/watch?v=8e4QB1DYYJU 4 https://www.youtube.com/watch?v=FpGGpgSEU94 5 https://www.youtube.com/watch?v=PFNe7NyJeb0 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 5 Understanding control Integrated engine control system - Secondary air Management - Evaporative emission canister purge - Automatic system adjustment - System diagnosis - Improvements in Electronic Engine Control system - Oxygen sensor improvements - Heated exhaust gas oxygen sensor - Fuel injection Timing - Transmission control - Automatic transmission control - Torque convertor lock up control - Differential and traction control Text / Reference: T/R Book Title/Author T1 U.Kienck, and L. Nielsen, Automotive Control Systems, SAE and Springer-Verlag 2000 LjuboVlacic, Michel Parent, Fumio Harashima, "Intelligent Vehicle Technologies", R1 Butterworth-Heinemann publications, Oxford, 2001 Crouse, W.H & Anglin, D.L,"Automotive

mechanics”, Intl.Student edition, R2 9th edition, TMH, New Delhi, 2002. William B. Ribbens-Understanding Automotive Electronics, 5th edition, Butterworth R3 Heinemann Woburn 1998. R4 Bosh, “Automotive Handbook”, 6th edition, SAE, 2004

Online Resources: Sl.No Website Link
 https://www.electronicsforu.com/technology-trends/tech-focus/automobile- 1 industry-sensor 2 https://www.youtube.com/watch?v=jyQuRgQHGCk 3 https://www.youtube.com/watch?v=8e4QB1DYYJU 4 https://www.youtube.com/watch?v=FpGGpgSEU94 5 https://www.youtube.com/watch?v=PFNe7NyJeb0
 definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

control		
M4.01	Explain integrated engine control systems	4
Understanding	Explain improvements in Electronic engine	
M4.02		5
Understanding	control system	
	Explain automatic transmission and traction	
M4.03		5
Understanding	control	
	Series Test – II	1

Contents:

Integrated engine control system - Secondary air Management - Evaporative emission canister purge - Automatic system adjustment - System diagnosis - Improvements in Electronic Engine Control system - Oxygen sensor improvements - Heated exhaust gas oxygen sensor - Fuel injection Timing - Transmission control - Automatic transmission control - Torque converter lock up control - Differential and traction control

Text / Reference:

T/R

Book Title/Author

T1 U.Kienck, and L. Nielsen, Automotive Control Systems, SAE and

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Outline the functions of engine and controls. (3 marks)

Answer: Begin with the standard definition or identity of *Outline the functions of engine and controls*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Outline the terms related to engine performance. (3 marks)

Answer: Begin with the standard definition or identity of *Outline the terms related to engine performance*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain the mapping of engine. 3 Understanding M1. 04 Outline electronic fuel control system 4 Understanding The concept of electronic Engine control system - Engine functions and control - Parameters and variables - Input to controller - output from controller - Definition of engine performance terms - Power - torque - fuel consumption - Volumetric efficiency - Thermal efficiency - calibration, Engine mapping-Effect of air fuel ratio, spark timing and exhaust gas recirculation on performance - Electronic fuel control system - Engine control sequence - open loop control - closed loop control - closed loop operation - Idle speed control. Identify various sensors and actuators used in electronic engine control. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the mapping of engine. 3 Understanding M1. 04 Outline electronic fuel control system 4 Understanding The concept of electronic Engine control system - Engine functions and control -*

Parameters and variables - Input to controller - output from controller - Definition of engine performance terms - Power - torque - fuel consumption - Volumetric efficiency - Thermal efficiency - calibration, Engine mapping-Effect of air fuel ratio, spark timing and exhaust gas recirculation on performance - Electronic fuel control system - Engine control sequence - open loop control - closed loop control - closed loop operation - Idle speed control. Identify various sensors and actuators used in electronic engine control. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 2

Define or explain Outline the engine Variables. (3 marks)

Answer: Begin with the standard definition or identity of *Outline the engine Variables*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Illustrate the working of various sensors. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate the working of various sensors*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Illustrate the working of various actuators. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate the working of various actuators*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Identify various actuators and actuators 1 Applying Typical electronic Engine Control System. Variables to be measured, Sensors and actuators- Temperature sensor- Pressure sensor-Mass air flow sensor-Oxygen sensor- Position sensor-Position/speed sensor-Knock sensor- Automotive Engine Control Actuators-Solenoid-switches-relays-motors-Fuel injection- Exhaust gas recirculation actuators-Variable valve timing.. (3 marks)

Answer: Begin with the standard definition or identity of *Identify various actuators and actuators 1 Applying Typical electronic Engine Control System. Variables to be measured, Sensors and actuators- Temperature sensor- Pressure sensor-Mass air flow sensor-Oxygen sensor- Position sensor-Position/speed sensor-Knock sensor- Automotive Engine Control Actuators-Solenoid-switches-relays-motors-Fuel injection- Exhaust gas recirculation actuators-Variable valve timing..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain 5 Understanding control systems.. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding control systems..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps,

application application application application.

Define or explain Explain various modes for fuel control. (3 marks)

Answer: Begin with the standard definition or identity of *Explain various modes for fuel control*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: application application definition, application principle/steps, application application application.

Define or explain Explain Electronic ignition control 5 Understanding Digital power train control systems - features - Components - Digital engine control system diagram - Modes for fuel control - Control during Engine crank - Engine Warm up - After the engine warm up - open loop - closed loop - Acceleration enrichment - Deceleration leaning - Idle speed control - EGR Control - Variable valve timing control - Electronic ignition control-Distributor less Ignition control. Outline the integrated engine control system and automatic transmission. (3 marks)

Answer: Begin with the standard definition or identity of *Explain Electronic ignition control 5 Understanding Digital power train control systems - features - Components - Digital engine control system diagram - Modes for fuel control - Control during Engine crank - Engine Warm up - After the engine warm up - open loop - closed loop - Acceleration enrichment - Deceleration leaning - Idle speed control - EGR Control - Variable valve timing control - Electronic ignition control-Distributor less Ignition control*. Outline the integrated engine control system and automatic transmission. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: application application definition, application principle/steps, application application application.

Module 4

Define or explain Explain integrated engine control systems 4 Understanding Explain improvements in Electronic engine. (3 marks)

Answer: Begin with the standard definition or identity of *Explain integrated engine control systems 4 Understanding Explain improvements in Electronic engine*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 5 Understanding control system Explain automatic transmission and traction. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding control system Explain automatic transmission and traction*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 5 Understanding control Integrated engine control system - Secondary air Management - Evaporative emission canister purge - Automatic system adjustment - System diagnosis - Improvements in Electronic Engine Control system - Oxygen sensor improvements - Heated exhaust gas oxygen sensor - Fuel injection Timing - Transmission control - Automatic transmission control - Torque convertor lock up control - Differential and traction control

Text / Reference: T/R Book Title/Author T1 U.Kienck, and L. Nielsen, *Automotive Control Systems*, SAE and Springer-Verlag 2000 LjuboVlacic, Michel Parent, Fumio Harashima, "Intelligent Vehicle Technologies", R1 Butterworth-Heinemann publications, Oxford, 2001 Crouse, W.H & Anglin, D.L, "Automotive mechanics", Intl.Student edition, R2 9th edition, TMH, New Delhi, 2002. William B. Ribbens-Understanding Automotive Electronics, 5th edition, Butter worth R3 Heinemann Woburn 1998. R4 Bosh, "Automotive Handbook", 6th edition, SAE, 2004 Online Resources: Sl.No Website Link <https://www.electronicsforu.com/technology-trends/tech-focus/automobile-1-industry-sensor-2>

<https://www.youtube.com/watch?v=jyQuRgQHGck> 3
<https://www.youtube.com/watch?v=8e4QB1DYYJU> 4
<https://www.youtube.com/watch?v=FpGGpgSEU94> 5
<https://www.youtube.com/watch?v=PFNe7NyJeb0>. (3 marks)

Answer: Begin with the standard definition or identity of 5 Understanding control Integrated engine control system - Secondary air Management - Evaporative emission canister purge - Automatic system adjustment - System diagnosis - Improvements in Electronic Engine Control system - Oxygen sensor improvements - Heated exhaust gas oxygen sensor - Fuel injection Timing - Transmission control - Automatic transmission control - Torque convertor lock up control - Differential and traction control Text / Reference: T/R Book Title/Author T1 U.Kienck, and L. Nielsen, Automotive Control Systems, SAE and Springer-Verlag 2000 LjuboVlacic, Michel Parent, Fumio Harashima, "Intelligent Vehicle Technologies", R1 Butterworth-Heinemann publications, Oxford, 2001 Crouse, W.H & Anglin, D.L, "Automotive mechanics", Intl.Student edition, R2 9th edition, TMH, New Delhi, 2002. William B. Ribbens-Understanding Automotive Electronics, 5th edition, Butter worth R3 Heinemann Woburn 1998. R4 Bosh, "Automotive Handbook", 6th edition, SAE, 2004 Online Resources: Sl.No Website Link <https://www.electronicsforu.com/technology-trends/tech-focus/automobile-1-industry-sensor> 2 <https://www.youtube.com/watch?v=jyQuRgQHGck> 3 <https://www.youtube.com/watch?v=8e4QB1DYYJU> 4 <https://www.youtube.com/watch?v=FpGGpgSEU94> 5 <https://www.youtube.com/watch?v=PFNe7NyJeb0>. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Outline the functions of engine and controls.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Outline the engine Variables.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **5 Understanding control systems..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Explain integrated engine control systems 4 Understanding Explain improvements in Electronic engine.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- <https://www.electronicsforu.com/technology-trends/tech-focus/automobile-industry-sensor> <https://www.youtube.com/watch?v=jyQuRgQHGCK>
<https://www.youtube.com/watch?v=8e4QB1DYYJU> <https://www.youtube.com/watch?v=FpGGpgSEU94> <https://www.youtube.com/watch?v=PFNe7NyJeb0>

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